2026 Senate forecast - Most_Magnificent_Model scheduled

Team 5: Elizabeth Miller, Mark Scott, Ethan Verrone, Boaz Alkalay, Leo Westebbe

Code and data: GitHub repository (https://github.com/MarkGScott960/2026_Midterm_Forecast) | This model's folder (https://github.com/MarkGScott960/2026_Midterm_Forecast/tree/main/Senate%20Forecast/Final%20Model)

scheduled run, 2026-10-30 14:10

P(DEMOCRATS >= 51 SEATS)

78.0%

simulation precision ±0.3 pts

EXPECTED DEMOCRATIC SEATS

52.4

median 52 | 80%: 49-56 | 95%: 47-57

KALSHI CONTROL MARKET

70.5%

sum of race prices: 52.4 seats

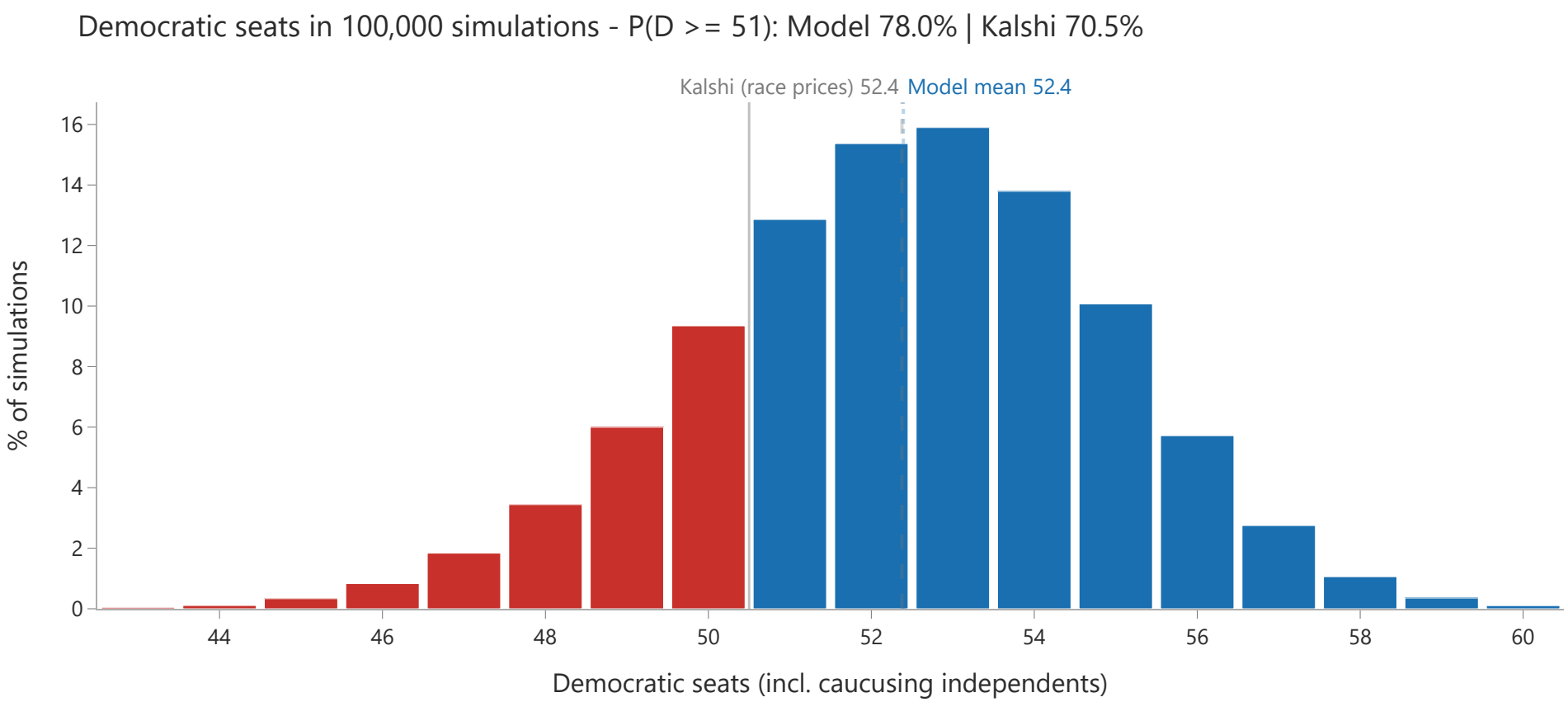
RUN

2026-10-30 14:10

100,000 simulations | data as of 2026-10-30 | 5 min

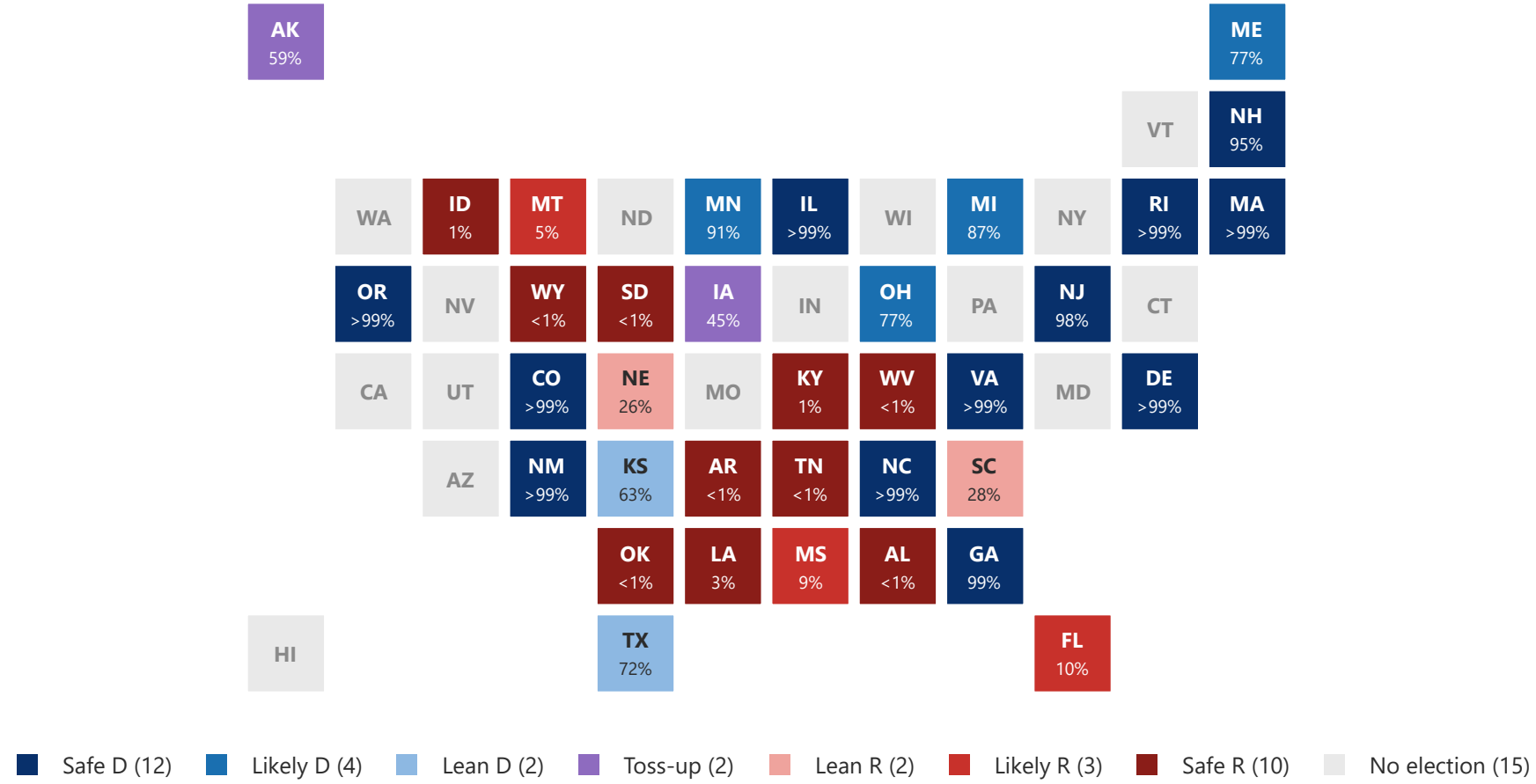
The model and Kalshi expect about the same number of Democratic seats (52.4 vs 52.4); the model's higher control odds come from rating Michigan, Kansas, Maine and Ohio higher for Democrats than the market does.

Seat distribution

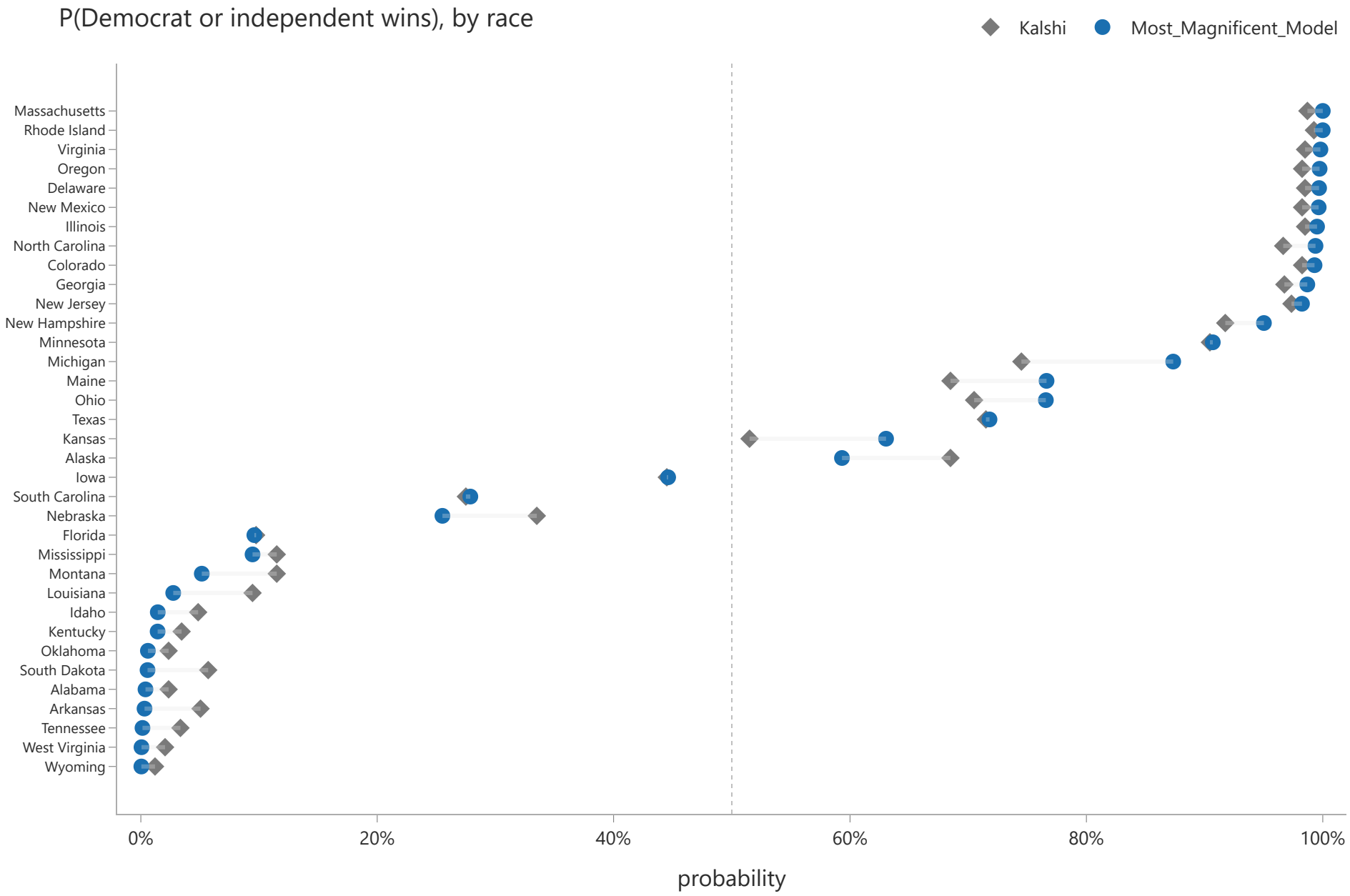


Map

Race ratings (number = Model P(D/Ind wins); D includes independents)
Toss-up under 60%, Lean 60%-75%, Likely 75%-95%, Safe 95%+



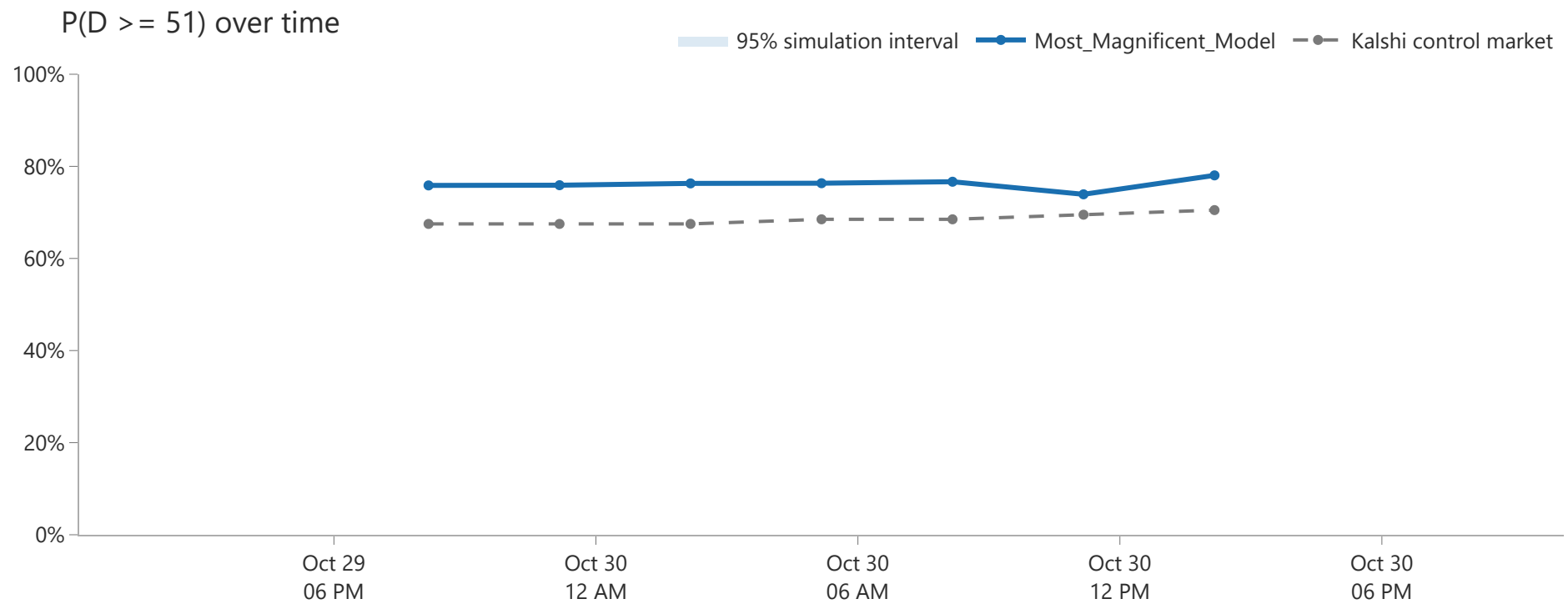
Race by race

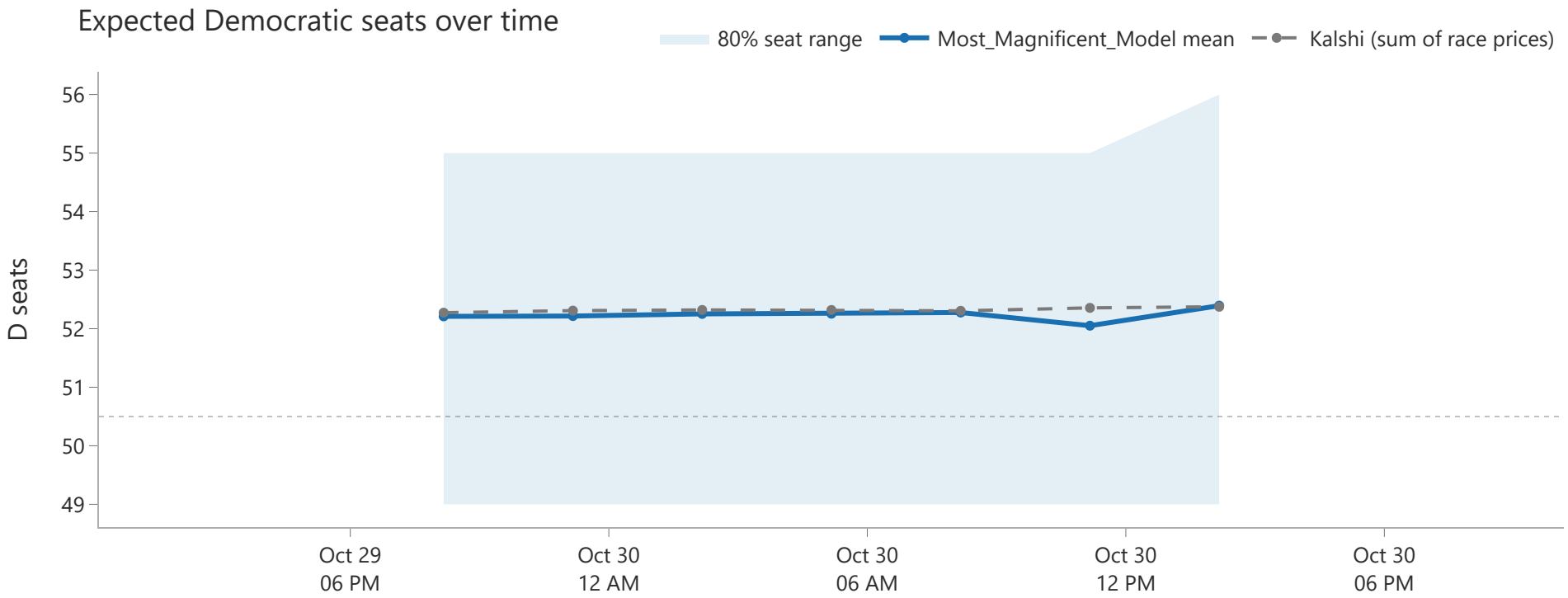

P(Democrat or independent wins), by race
Kalshi
Most_Magnificent_Model
Massachusetts
Rhode Island
Virginia
Oregon
Delaware
New Mexico
Illinois
North Carolina
Colorado
Georgia
New Jersey
New Hampshire
Minnesota
Michigan
Maine
Ohio
Texas
Kansas
Alaska
Iowa
South Carolina
Nebraska
Florida
Mississippi
Montana
Louisiana
Idaho
Kentucky
Oklahoma
South Dakota
Alabama
Arkansas
Tennessee
West Virginia
Wyoming
0%
20%
40%
60%
80%
100%
probability

Race	Call	Model P(D/Ind)	Kalshi P(D/Ind)	Model - Kalshi (pts)	Projected margin	+/- (1 SD)	Polls	Model change since first run (pts)	Kalshi change since first run (pts)
Massachusetts	Safe D	>99%	99%	+1	D+34.2	+/-5.7	13	+0	+0
Rhode Island	Safe D	>99%	>99%	+1	D+25.3	+/-5.5	4	+0	+0
Virginia	Safe D	>99%	98%	+1	D+18.7	+/-5.9	1	-0	+0
Oregon	Safe D	>99%	98%	+1	D+28.5	+/-9.4	0	+0	-1
Delaware	Safe D	>99%	98%	+1	D+28.5	+/-9.4	0	-0	+0
New Mexico	Safe D	>99%	98%	+1	D+18.7	+/-6.4	2	-0	-0
Illinois	Safe D	>99%	98%	+1	D+14.6	+/-5.4	3	-0	+0
North Carolina	Safe D	>99%	97%	+3	D+12.3	+/-4.7	40	+0	+0
Colorado	Safe D	>99%	98%	+1	D+24.5	+/-9.4	0	-0	+0
Georgia	Safe D	99%	97%	+2	D+10.9	+/-4.8	18	+0	+1
New Jersey	Safe D	98%	97%	+1	D+20.5	+/-9.4	0	-0	-0
New Hampshire	Safe D	95%	92%	+3	D+8.3	+/-5.1	21	+0	+1
Minnesota	Likely D	91%	90%	+0	D+6.3	+/-4.9	7	+0	+1
Michigan	Likely D	87%	74%	+13	D+5.2	+/-4.7	43	-0	+0
Maine	Likely D	77%	68%	+8	D+3.2	+/-4.6	18	+1	+5
Ohio	Likely D	77%	70%	+6	D+3.2	+/-4.6	32	+0	+1
Texas	Lean D	72%	72%	+0	D+2.4	+/-4.5	53	+2	+3
Kansas	Lean D	63%	52%	+12	D+1.5	+/-4.6	7	+15	+3
Alaska	Toss-up	59%	68%	-9	D+1.0	+/-4.7	18	-0	+0
Iowa	Toss-up	45%	44%	+0	R+0.6	+/-4.7	24	-1	-3
South Carolina	Lean R	28%	28%	+0	R+2.7	+/-4.7	4	-1	+0
Nebraska	Lean R	26%	33%	-8	R+3.3	+/-5.1	4	+0	+1

Race	Call	Model P(D/Ind)	Kalshi P(D/Ind)	Model - Kalshi (pts)	Projected margin	+/- (1 SD)	Polls	Model change since first run (pts)	Kalshi change since first run (pts)
Florida	Likely R	10%	10%	-0	R+6.0	+/-4.6	17	-0	-1
Mississippi	Likely R	9%	11%	-2	R+7.2	+/-5.5	3	+0	+0
Montana	Likely R	5%	11%	-6	R+9.4	+/-5.8	26	+1	-1
Louisiana	Safe R	3%	9%	-7	R+13.7	+/-7.1	0	-0	-0
Idaho	Safe R	1%	5%	-3	R+15.4	+/-6.9	1	-0	-1
Kentucky	Safe R	1%	3%	-2	R+21.3	+/-9.4	0	-0	+0
Oklahoma	Safe R	<1%	2%	-2	R+25.3	+/-9.4	0	-0	-0
South Dakota	Safe R	<1%	6%	-5	R+13.8	+/-5.2	4	+0	+0
Alabama	Safe R	<1%	2%	-2	R+16.3	+/-5.8	5	-0	+0
Arkansas	Safe R	<1%	5%	-5	R+14.5	+/-5.0	5	-0	+0
Tennessee	Safe R	<1%	3%	-3	R+23.0	+/-6.9	1	-0	-0
West Virginia	Safe R	<1%	2%	-2	R+37.1	+/-9.4	0	-0	+0
Wyoming	Safe R	<1%	1%	-1	R+37.3	+/-9.4	0	-0	+0

History





Notes

- Model Testing: 35 PASS | 5 WARN | 0 FAIL (details in model_warnings.xlsx).
- Data: votehub_senate live, votehub_generic_ballot live, votehub_approval live, kalshi_races live, kalshi_national live.

Key settings

Poll source	VoteHub API
Poll age	all polls, half-life 7 days
Polls : fundamentals : Kalshi	Bayesian precision (poll floor 3.9, prior SD 9.3); Kalshi precision 1/8^2, longshot-corrected, conversion SD >= 4.42
Poll reliability (thin races)	automatic (Bayesian precision)
Fundamentals	PVI x 2 + national environment (generic-ballot blend 0.75) + candidate effects x 0.30
Incumbency	3.8 pts (appointed 1)
LV : RV : A	1 : 0.85 : 0.7; RV shift +0.9 (measured)
Flooding penalty	1/k^0.5
House effects	in-cycle, shrunk to scorecard (k 5)
Ungraded pollsters	random
National signals	VoteHub polls + Kalshi averages (= 5 polls)
Kalshi races	Bayesian input + comparison (found automatically)
Source errors	correlated (rho 0.3); sims: national 2.5 + region 1.5 + state, in pts
Calibration	calibration.json (2026-10-29 22:45)
Simulations	100,000

Equations and weights

1. WEIGHT OF EACH POLL

$$w = 0.5^{\text{age}/7} \times \left(\frac{\min(n,\,2500)}{800}\right)^{0.5} \times w_{\text{pop}} \times w_{\text{partisan}} \times w_{\text{generic}} \times w_{\text{field}} \times w_{\text{quality}} \times w_{\text{accuracy}} \times \frac{1}{k^{0.5}} \times \frac{1}{\text{versions}}$$

Population: LV 1, RV 0.85, adults 0.7. Partisan sponsor 0.6. Generic-party question 0.5. Full field 0.9. Quality 0.5 to 1 by VoteHub grade (ungraded firms get a random graded firm's grade in each simulation). Accuracy = (1.25 x 4 / poll SD)^2, kept between 0.5 and 2. k = the pollster's polls of that race.

2. EACH POLL'S MARGIN

$$m = (D - R)\,\frac{D + R + U}{D + R} + s_{RV} \pm s_{\text{partisan}} - h, \qquad h = \frac{\sum \text{residuals} + 5\,h_{\text{scorecard}}}{n_{\text{polls}} + 5}$$

U = undecided. RV shift s_RV = +0.88 pts (registered-voter polls only; measured from LV/RV pairs). Partisan shift = 3 pts against the sponsor. Residual = poll minus the same state's average of other pollsters, from the last 60 days.

3. POLLING AVERAGE

$$\bar{m} = \frac{\sum w\,m}{\sum w}, \qquad SD_{\text{polls}} = \sqrt{\frac{SD_{\text{poll}}^2}{n_{\text{eff}}} + 3.87^2 + SD_U^2}, \qquad n_{\text{eff}} = \frac{(\sum w)^2}{\sum w^2}$$

The middle term is the error all polls of a state share, so more polls can't remove it.

---

4. NATIONAL ENVIRONMENT

$$\text{env} = 0.75 \times \text{GB} + 0.25 \times 0.35 \times (-\text{net approval}), \qquad \text{signal} = \frac{n_{\text{fresh}} \cdot \text{VoteHub} + 5 \cdot \text{Kalshi}}{n_{\text{fresh}} + 5}$$

GB = generic ballot. Each VoteHub poll counts 0.5^(age / 14) toward n_fresh. This run: env = D+8.7.

---

5. PRIOR (FUNDAMENTALS)

$$\text{prior} = 2 \times \text{PVI} + \text{env} \pm 3.78\,[\text{incumbent}] + 0.30 \times c, \qquad SD_{\text{prior}} = 9.32$$

Appointed incumbents get 1 pt. c = candidate effect: the nominees' past margin minus a generic candidate's (capped at +/- 25). Independent-led races: SD x 1.3.

---

6. KALSHI

$$P' = \text{logistic}\,(1.15 \cdot \text{logit}\,P), \qquad \text{margin}_K = \Phi^{-1}(P') \cdot \max\,(4.42,\ SD_{F+P}), \qquad SD_K = 8$$

Used only when 6% <= P <= 94%.

---

7. EACH RACE

$$\text{margin} = \frac{\pi_F \cdot \text{prior} + \pi_P \cdot \bar{m} + \pi_K \cdot \text{margin}_K}{\pi_F + \pi_P + \pi_K}, \quad \pi = \frac{1}{SD^2}, \quad SD^2 = \frac{1}{\Pi} + \frac{2(0.3)}{\Pi^2}\left(\sqrt{\pi_F\pi_P} + \sqrt{\pi_F\pi_K} + \sqrt{\pi_P\pi_K}\right) + 0.76^2 \cdot \text{days left}$$

Pi = the sum of the three precisions. The second term widens the range because the three sources tend to miss together.

---

8. SIMULATIONS

$$\tilde{m}_i = \text{margin}_i + N + R_{g(i)} + L\,f_i + e_i, \qquad \text{D seats} = 34 + \sum_i \mathbf{1}[\tilde{m}_i > 0], \qquad P(D \geq 51) = \text{share of simulations}$$

N = national miss (SD 2.52, t with df 8). R = regional miss (SD 1.51). L = lean factor (SD 1), f_i = PVI / 10. e = the rest of the race's SD (t with df 10.9). An independent's win counts for Democrats with probability: Nebraska 0.8.
